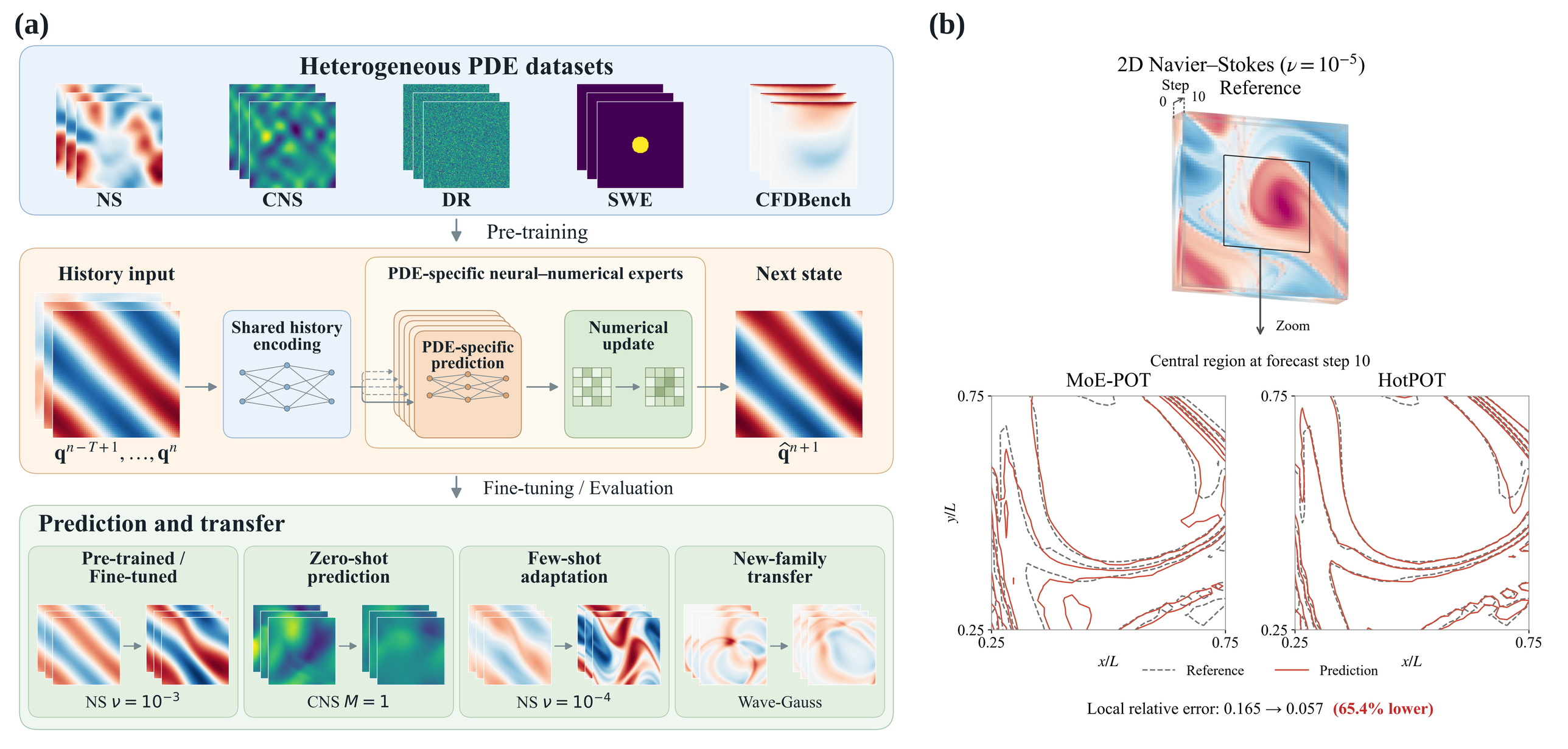

(a)
Heterogeneous PDE datasets
NS
CNS
DR
SWE
CFDBench
Pre-training
History input
$\mathbf{q}^{n-T+1}, \ldots, \mathbf{q}^{n}$
Shared history encoding
PDE-specific neural–numerical experts
PDE-specific prediction
Numerical update
Next state
$\widehat{\mathbf{q}}^{n+1}$
Fine-tuning / Evaluation
Prediction and transfer
Pre-trained / Fine-tuned
NS $\nu = 10^{-3}$
Zero-shot prediction
CNS $M = 1$
Few-shot adaptation
NS $\nu = 10^{-4}$
New-family transfer
Wave-Gauss
(b)
2D Navier–Stokes ($\nu = 10^{-5}$)
Step
0
10
Reference
Zoom
Central region at forecast step 10
MoE-POT
HotPOT
0.75
0.25
0.25
0.75
y/L
x/L
Reference
Prediction
Local relative error: 0.165 → 0.057 (65.4% lower)